

Fig. 4: Actual-size, single-side PCB for the Arduino-based tilt detector

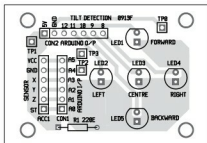


Fig. 5: Component layout for the PCB

PARTS LIST

Semiconductors:

ACC1 - Accelerometer module

LED1-LED5 - 5mm LED

Resistor (1/4-watt, $\pm 5\%$ carbon):

R1 - 220-ohm

Miscellaneous:

CON1 - 6-pin connector

CON2 - 7-pin connector

Board1 - Arduino Uno board

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download on www.efymag.com website.

the central LED (red) will glow. Pins 8 through 12 of Board1 are connected to LEDs as shown in Fig. 3.

Construction and testing

Actual-size, single-side PCB for the Arduino-based tilt detector is shown in Fig. 4 and corresponding component side layout is shown in Fig. 5. Suitable connectors are provided on the PCB to interface accelerometer and Arduino Uno board. To test the circuit, check the correct voltage levels as shown in the test points table.

For functional testing follow below-mentioned steps:

1. After assembling the circuit, first

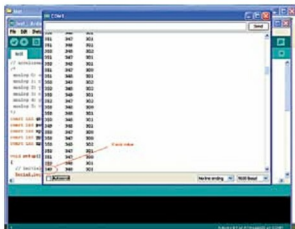


Fig. 6: Screenshot of values of X, Y and Z axes

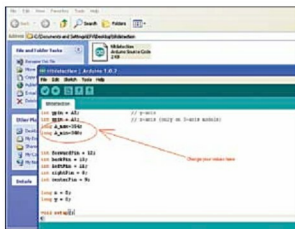


Fig. 7: Screenshot of Arduino Uno programming

we need to trigger values for LEDs, and for that:

- (i) Connect the Arduino board to your computer with USB cable.
- (ii) Download the source code (test.ino) and burn it on the Arduino board.
- (iii) Open the serial port from Arduino environment and press reset button.
- (iv) You will see the screen shown in Fig. 6 on which the values of X, Y, and Z axes will vary corresponding to the position of your accelerometer.
- (v) Tilt it to the left and note the value for 'A_max' after which you want the corresponding LED to glow. The same trigger will be applied for backward tilt too.
- (vi) Tilt it to the right and note the value for 'A_min' after which you want the corresponding LED to glow. The same trigger will be applied for forward tilt too.

2. The trigger values are defined by 'A_max' and 'A_min' in the final code (tiltdetection.ino).

3. Download the final code and open it in Arduino IDE.

4. Put the values for 'A_max' and 'A_min' in the final code as shown in Fig. 7.

5. Program Arduino again with this new code and the system is ready to work.

6. If the angular tilt is within the threshold, red LED will glow. Otherwise LEDs corresponding to each direction will glow.

7. You can see the same on serial port also. Connect the USB and open the serial port in Arduino environment. You will see on the serial port 'forward' if tilted forward, and similarly for the other directions.

Software

The software is written in Arduino programming language and the board is programmed using Arduino IDE software. ATmega328 on Arduino Uno comes pre-burnt with a bootloader that allows you to upload new code without the use of external hardware programmer. It communicates using the original STK500 protocol. You can also bypass the bootloader and program the microcontroller through ICSP (in-circuit serial programming) header, but using bootloader is quicker and easier. Select Arduino Uno from the 'Tools > Board' menu (according to the microcontroller on your board) in the Arduino IDE and burn the program through standard USB port in the computer. ●

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